

TrES-3b

R-2749

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_210707 · created 2026-09-02T23:06:38+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459402.819 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.054
Transit depth (Rp/Rs)^2	3.86 +/- 2.11 [%]
Orbital Inclination (inc)	82.87 +/- 2.29
Airmass coefficient 1 (a1)	0.87 +/- 0.1
Airmass coefficient 2 (a2)	0.13 +/- 0.11
Scatter in the residuals of the lightcurve fit is	11.4 %
Best Comparison Star	#1 - [604, 421]
Optimal Method	PSF photometry
Transit Duration (day)	0.061 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.196 $\pm$ 0.054 vs 0.16309 (deviation 0.35 σ)
Duration fit vs published	0.061 d vs 0.0567 d (deviation 0.12 σ)
Mid-time O-C	-1.0 min
Depth SNR	2.1
Residual scatter	11.4 %

Light curve

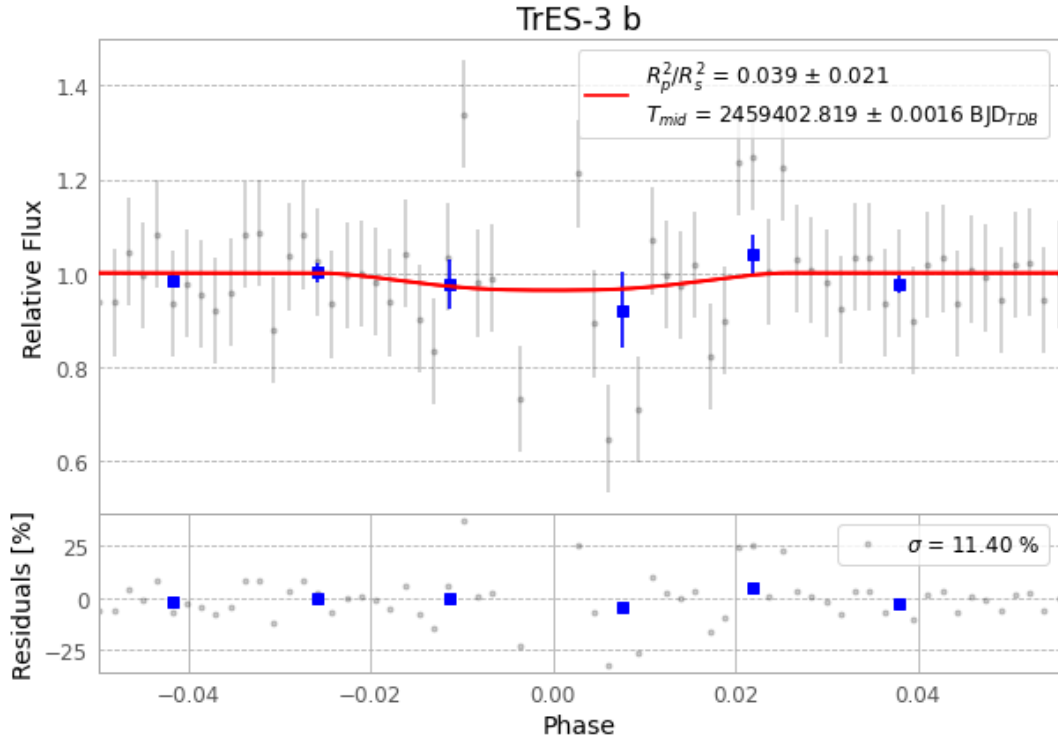


Figure 1 — FinalLightCurve_TrES-3 b_2021-07-06.png (SHA-256 42040186e5f7712e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [363, 205], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c9d5b177c99d7cf8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), TRES-3210707060012.FITS → TRES-3210707091812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-210707.log (0151f17c0cd9...), FinalLightCurve_TrES-3 b_2021-07-06.png (42040186e5f7...), FinalLightCurve_TrES-3 b_2021-07-06.csv (c15eda335118...)

Compute resources

Wall time 162.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 23:06Z.